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EDUCATION

UC San Diego

Sep 2021 - Dec 2025

Bachelor of Science in Mechanical Engineering

TECHNICAL SKILLS

- **Fabrication:** Design for manufacturing, CNC milling, lathe machining, welding, 3D printing, soldering, DC circuit design and assembly
- **Software:** SolidWorks, Fusion 360, ANSYS, AutoCAD, Microsoft Suite, KiCad, DaVinci Resolve
- **Programming:** MATLAB, Arduino C++, LabVIEW, HTML, Python
- **Interpersonal:** Leadership in team-based environments, attention to detail, conflict resolution, project management tools, written & oral communication

PROJECT EXPERIENCE

Triton Solar Car — Engineering Project Team at UC San Diego

Oct 2021 - Jun 2025

Project Lead

- Ensured proper communication between business partners and team members on a team of 15+ members.
- Oversaw the technical development of our solar powered car on the mechanical and electrical teams.
- Performed administrative duties such as managing the leadership on the team, creating project schedules, and faculty communication.
- Modeled car chassis in Fusion 360 and performed impact force simulations in ANSYS.

Business Lead

- Secured over \$10,000 in grant funding through detailed grant proposals.
- Managed finances relating to Triton Solar Car.
- Networked with local businesses and corporations to gain funding, expertise, and services for the team.
- Marketed the team to the student body to garner interest in the team and its values.

Cosmic Ray Deflection System

Feb 2026 - Present

- Investigating alternative methods for long-term cosmic ray protection for long-term space flights.
- Creating MATLAB simulations for aiding in design for proposed deflection system.
- Researching current methods and key faults of current shielding from cosmic rays on spacecraft.
- Developing knowledge of nuclear physics, materials science, and electromagnetism as needed during investigation.

PCB-Wound Electromagnet Design Automation

Feb 2026 - Present

- Created a program to automatically design and simulate PCB-wound electromagnets with set parameters.
- Created MATLAB script to model generated magnetic field for electromagnets in a given count and arrangement.
- Enabled program to export electromagnet design data to import to KiCad for manufacturing.
- Planning to order a set of program-designed PCBs for further testing for future project.

Haptic Feedback Prosthetic Acclimation System

Jan 2026 - Present

- Programmed microcontroller to interpret gravitational data and provide appropriate haptic feedback for a given missing limb.
- Assembled electronics required for motor-based vibrational haptic feedback.
- Designing ergonomic housing for all electronics for limb-mounted design as proof of concept.

Spectrometer pH Measurement Testbench

Sep 2025 - Dec 2025

- Designed automated testbench for seawater pH and alkalinity calibration minimizing vibration, temperature drift, and light interference.
- Prepared deliverables and technical documentation on a weekly basis to professor Todd Martz.
- Designed fiber optic testbench in Fusion 360 with appropriate GD&T and minimal use of 3D printed supports when printed.
- Managed team deadlines and objectives for a team of three.
- Modified existing LabView programs to fully automate titration and measurement of pH and alkalinity of given samples.

Regenerative Braking Trailer – Senior Design Project

Apr 2025 - Jun 2025

- Designed and built a hitch-mounted trailer capable of generating electrical energy during vehicle braking.
- Modeled mechanical components in Fusion 360 and performed static load analysis to ensure structural integrity.
- Developed ESP32-based control system including PCB design, firmware, and sensor integration.
- Created comprehensive BOM, test procedures, and weekly technical documentation.
- Executed full-scale braking tests using high-power safety protocols.

Custom Electric Moped Project

Apr 2023 - Present

- Worked to design and fabricate a long-range electric moped with auxiliary power and storage for recreational use.
- Performed finite element analysis in Fusion 360 for a maximum rider weight capacity of 250 pounds.
- Designed, assembled, and tested electrical system with high-power safety measures in mind.
- Laser cut sheet metal to assemble into chassis.

Robot Arm Project – MAE 3 Class Project

Jan 2023 - Mar 2023

- Worked on a team to design and built a 3-DOF remote-controlled robotic arm for a timed competition task.
- Prepared deliverables and technical documentation on a weekly basis to professor Nathan Delson.
- Modeled the robotic arm in Fusion 360, modeling with appropriate GD&T. Used AutoCAD to prepare and laser cut acrylic for various components of the robot.

WORK EXPERIENCE

Line Cook — JOEY La Jolla

Mar 2026 - Present

- Prep all ingredients and food products to fulfill prep lists, par levels, and recipe standards.
- Follow all instructions and guidelines for safe usage of kitchen equipment and cleaning compounds.
- Clean, organize, and maintain refrigerators, cooler drawers, ovens, prep and dry storage areas, etc.
- Complete assigned side work and closing duties to set your team up for success. Stock, clean, and organize line stations.

Camp Counselor — Knock Around Summer Camp

Jun 2022 - Aug 2025

- Worked as a day camp counselor with kids aged 5-13.
- Facilitated various activities, games, and fostered appropriate behavior.
- Managed behavioral conflicts between campers.
- Emphasized not disturbing student life while camp is in session.

Camp Coordinator — Knock Around Summer Camp

Jun 2024 - Aug 2024

- Led daily rock climbing sessions with groups of 15+ campers, ensuring use of proper safety protocol.
- Executed pre-planned activities in collaboration with other counselors.
- Acted as a direct point of contact for other counselors.
- Utilized adaptability in the event of emergency response.

Store Manager — The Landing

Jan 2020 - May 2020

- Prepared and served specialty beverages with speed and consistency.
- Operated Square POS and handled transactions accurately.
- Greeted customers entering the store and fostered positive atmosphere.
- Handled inventory and cleaned equipment as needed.